

Appreciating Neuron Morphology using Golgi-Cox Staining on Rat Brain Tissue

Yatin Prakash^{1*}

¹ Indian Institute of Science Education and Research, Bhopal

ABSTRACT

Neuronal morphology-particularly the geometry of dendritic arborizations and spine distributions is the physical basis of the formation of circuits that enable higher-order functions. We used the NCBS Golgi-Cox protocol on adult rat brain tissue to visualize neuron architecture across the cortex and hippocampus. Brains were impregnated in a mercuric chloride/potassium dichromate/chromate solution, vibratome-sectioned at 150 μ m, and developed by alkaline reduction to yield an insoluble intracellular mercury precipitate. Using brightfield microscopy, we could unambiguously trace and see isolated, darkly stained pyramidal, stellate, and interneurons against a low-contrast background field. Dendritic spines were resolved at high magnification. Regional variation in impregnation density was observed throughout, most notably in the hippocampus, where it was dense.

Keywords: Neuron Morphology, Golgi-Cox Staining

1. INTRODUCTION

Neuronal morphology has to be the most fascinating feature of the nervous system. It is this highly variable and complex structure, and its dynamic nature, that allows it to perform the multitude of brilliant functions that the organism can execute. Neurons operate with an unimaginable amount of efficiency and precision while receiving and sending feedback from multiple other neurons, all with a minimal power consumption of about 20W. Compare this to basic everyday items like lightbulbs and chargers that consume triple or quadruple that amount.

The function of a neuron is inseparable from its geometry: parameters like dendritic arborization and spine density are physical correlates of synaptic weight and connectivity, so any technique that reveals the geometry is effectively a readout of a circuit's wiring logic. It is also useful to understand pathomechanisms of disease progression and severity in clinical settings. It can be used to explore computational and information processing capabilities in both single neurons and across established connected circuits. Learning about their connectivity, topology, and plasticity at the neuron and further systems levels can be then applied to neuromorphic computing to engineer brain-inspired hardware and software.

Three major Golgi staining subtypes exist: Rapid Golgi, Golgi-Kopsch, and Golgi-Cox. Of these, the Golgi-Cox method is considered the most reliable for demonstrating dendritic arborization with a low background. It also doesn't require the temperature-sensitive osmic acid step that makes the rapid Golgi unpredictable. Many modifications of this method have been conducted and commercial kits have been developed and optimised; most aimed to increase its reliability, reduce the required time, and to increase the selectivity of neuronal vs. glial cells. (Koyama & Tohyama, 2012) (Zaqout & Kaindl, 2016)

Protocols have been developed at both NIMHANS and NCBS, primarily to further optimize the existing protocols for their purposes. In this project we used the protocol from NCBS to visualize rat brain tissue.

In this project, I got hands-on with Golgi-Cox staining and learned animal handling, brain tissue processing, sectioning, and visualizing using an adult rat model to characterize neuronal morphology across the cortex and hippocampus.

2. METHODOLOGY

General Instructions Before Starting

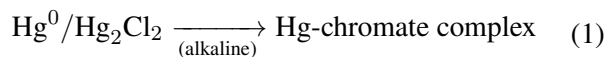
1. All glass and plastic bottles should be rinsed with fresh double distilled water (dd-H₂O) before use.

2. Metal instruments must be avoided during the impregnation step.
3. The amount of solutions and gelatin-coated slides should be prepared according to the number of samples as indicated in each section.
4. The incubation time frames given for various steps in this protocol are periods that do not affect significantly the staining quality. Thus, these time frames provide some flexibility when pursuing with the protocol.
5. All solutions must be stored in the dark using aluminum foil in a covered, lightproof box.
6. Care should be taken with all solutions due to toxicity and carcinogenesis: direct skin contact and inhalation can be avoided by wearing gloves and performing the experiments under a chemical hood, respectively.
7. Waste should be handled with care and disposed off along with hazardous wastes in secured bio-hazard canisters.

3. METHODOLOGY

3.1. Theoretical Basis

Golgi-Cox impregnation relies on the in situ formation of an insoluble mercury dichromate/chromate complex inside a small, stochastically determined subset of neurons. The mercuric ion (Hg^{2+}) is the species ultimately converted to a visible, insoluble reduction product. This is a schematic reduction pathway and is not fully resolved in the literature and is treated here schematically:



This is why impregnated tissue shows no colour change during the 1-3 week impregnation phase itself; the deposited complex only becomes visible after black reaction.

3.2. Solution Preparation

Preparation of Golgi-Cox solution for rat brain on (12/06/2026)

- **Stock Ratio:** Mixed $\text{K}_2\text{Cr}_2\text{O}_7$: HgCl_2 : K_2CrO_4 : ddH_2O in a 5 : 5 : 4 : 10 ratio (= 24 parts total).
- **Main Batch (7 Brains):** Scaled up for 7 brains requiring 192 mL total:

$$192 \text{ mL} \div 24 = 8 \text{ mL/part}$$

- 40 mL $\text{K}_2\text{Cr}_2\text{O}_7$ (2 g)
- 40 mL HgCl_2 (2 g)

- 32 mL K_2CrO_4 (1.6 g)
- 80 mL Distilled Water (DW)

3.3. Preparation of Brain Tissue

Rats were taken from their cages to the dissection room. They were anesthetized using isoflurane inside a closed chamber. Cervical dislocation was performed, and the rats were decapitated using a guillotine. Brains were carefully cut-off from the regions it was still connected, shelled out and kept in freshly prepared Golgi-Cox solution. The entire procedure was done very rapidly to maintain the physiological conditions of neuronal tissues. Brains were kept in amber-colored falcon tubes (containing solution C) for Dipped brains in the solution, kept for 7 days in the dark (till 19/06/26)

3.4. Sectioning of Brain Tissues

After 8-10 days, sections were taken using a vibratome. A total of 6 g of sucrose was dissolved in 100 mL of distilled water to make a 6% sucrose solution, which served as the sectioning medium.

The brain was fixed onto the stage using glue (Fevikwik) and placed inside a sectioning tray filled with the 6% sucrose solution. 150 μm thick sections were collected on 3% gelatin-coated slides.

3.5. Tissue Processing

- i. The sections were gently pressed after being placed on the slides, using blotting paper so that the sections adhered to the slides.
- ii. Sections were then kept in a moist chamber for 1 h. The sections were then washed with distilled water 2-3 times, to clear out excess sucrose.
- iii. **Alkali treatment (black reaction):** 5% sodium carbonate (Na_2CO_3) was prepared by dissolving 5–10 g of sodium carbonate in 100 mL of distilled water, with a pH of 10–11. This solution was poured over the sections and kept at room temperature for approximately 55 minutes in the dark.

Note: It should be ensured that the place is completely dark; otherwise, the reaction will not take place. (We kept the slides in the instrumentation room). At this stage: The complex is not itself coloured- the yellow-orange hue of the stock and impregnation bath comes from the $\text{K}_2\text{Cr}_2\text{O}_7$ and K_2CrO_4 ions in solution

- iv. Sections were washed again with distilled water to remove excess Na_2CO_3 .
- v. **Dehydration:** Slides were then transferred into graded alcohol (ethanol): 70% ethanol for 12 min

Compound	Concentration	Preparation
1. Potassium dichromate (K ₂ Cr ₂ O ₇)	5% (5 g in 100 ml distilled water)	1 was poured over 2 and mixed thoroughly to give Solution A . Note: Beaker wrapped with aluminium foil throughout; solution is light-sensitive.
2. Mercuric chloride (HgCl ₂)	5% (5 g in 100 ml distilled water)	
3. Potassium chromate (K ₂ CrO ₄)	4% (3.2 g in 80 ml distilled water)	3 and 4 were mixed to give Solution B .
4. Distilled water	200 ml	
Solution A was added slowly to Solution B in small aliquots under dark conditions with continuous mixing on a magnetic stirrer to give Solution C (Golgi–Cox solution). pH=6.5. Note: A dark reddish-yellow deposit formed upon mixing; Solution C must be free of this deposit before brains are introduced. The solution was left for 2 h until no further deposits formed.		
Solution C was filtered through Whatman filter paper and stored in amber falcon tubes in a dark place. Note: Golgi–Cox solution should be freshly prepared immediately before euthanising the animals. Volumes here were scaled from the original protocol to accommodate demonstration to multiple observers.		

Table 1. Preparation of Golgi–Cox impregnation solution.

then 80% ethanol for 7 min then 90% ethanol for 7 min then in 100% ethanol we dip it quickly and remove it.

- vi. **Mounting:** Sections were dipped in Xylene and mounted with D.P.X. (D.P.X is Di-butyl Pthalate, which is a plasticiser and helps keep dried resin flexible; Polystyrene is a synthetic resin that helps physically lock the tissue section in place; Xylene is a solvent helps keep plasticiser and resin in liquid state and itself evaporates quickly)

3.6. Miscellaneous

When preparing the slides dip them in 1% HCl overnight and Coat them with gelatin.

Prepared Gelatin using 200ml ddH₂O, 4g Gelatin, and 0.5g of Chrom Alum

Prepared Na₃PO₃ buffer as (Monobasic- 1.48g, Dibasic- 6.76g) Sucrose(28.8g) and 480ml of ddH₂O. For storage, Prepared PB buffer + 6% sucrose (200 ml PB, 0.1M, + 6% sucrose)

Types of Golgi-Staining:Find in table below

3.7.

NOTE: We were informed that the rats were a part of a older study that concerned autism. We did not know whether the rats were autistic or not to begin with.

4. RESULTS

Golgi–Cox impregnation of adult rat brain tissue produced the characteristic sparse labeling pattern—approximately 1–3% of neurons filled across all sampled regions. The selectivity of labeling was sufficient in most fields to permit unambiguous single-cell tracing. Background precipitation, where present, was low and punctate rather than diffuse, indicating that the alkaline development was complete. A diffuse background with random precipitation would have indicated premature light exposure during impregnation.

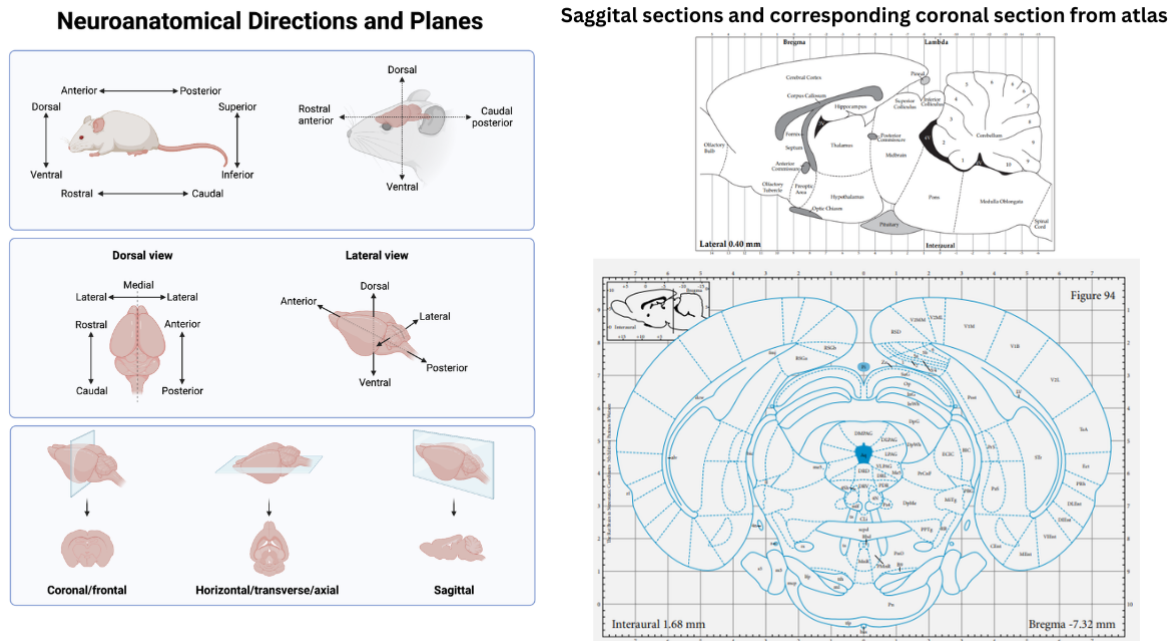
Brightfield imaging using an epifluorescence microscope was used at three magnifications: low power to assess regional distribution of impregnated neurons across cortex, hippocampus and cerebellum. Dendritic spines can also be visualized in much higher magnification along both secondary/tertiary dendrites.

We referred to this atlas to see if our sections somewhat correlated to some defined regions. (Paxinos & Watson, 2004)

Image-J, Fiji was used to post-process the images locally from Olympus cellSense. The individual neurons seemed to be well stained darkly black against a slightly orange background. The sections were slightly tattered and discontinuous, probably due

Table 2. Comparison of the three classical Golgi impregnation methods:

Method	Reagent sequence	Deposited species	Visualization timing	Rate-limiting sensitivity	Primary use-case
Rapid Golgi	OsO_4 + $\text{K}_2\text{Cr}_2\text{O}_7$ mordant $\rightarrow \text{AgNO}_3$	Metallic Ag	Immediate, during impregnation (days)	Temperature (osmication step)	Finest resolution (single-cell spines, thin collaterals)
Golgi-Kopsch	HCHO /glutaraldehyde pre-fix $\rightarrow \text{K}_2\text{Cr}_2\text{O}_7$ + AgNO_3	Silver chromate	Immediate, during impregnation	Baseline fixation quality of source tissue	Autopsy/archival human tissue tolerant of poor starting quality
Golgi-Cox	HgCl_2 + $\text{K}_2\text{Cr}_2\text{O}_7$ + $\text{K}_2\text{CrO}_4 \rightarrow \text{Hg-complex}$ (no AgNO_3)	Mercury-chromate complex	Delayed; only after alkaline (NH_3) reduction post-sectioning	Time (1 to 3 weeks), largely temperature independent	Reproducible, quantitative spine-density comparison across animals and groups

**Figure 1.** Dissection planes and correlating sections from atlas

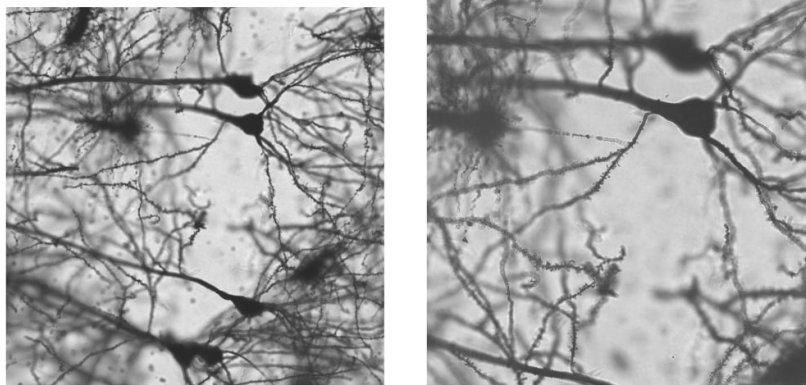
to excessive time spent in high concentration ethanol during the dehydration step or due to Xylene during the DPX step. It could also be a vibratome shearing artifact likely resulting from suboptimal blade speed or disintegration of the structure during any other step.

We were able to find various kinds of neurons. In the cortex regions we could clearly distinguish the layered organisation of the cortex. There were pyramidal neurons (identified by their triangular cyton), tiny stellate neurons, and some interneurons. Apical and basal dendritic trees were traceable from the soma. In the hippocampus the neurons were relatively bunched up and clustered compared to the cortex, here too there were pyramidal neurons. Refer attached micrographs for detailed results.

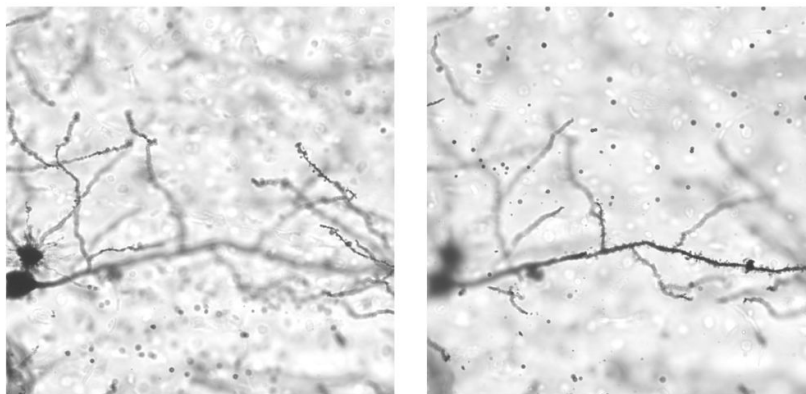
5. DISCUSSION

5.1. Why only 1–3% of neurons take up the stain?

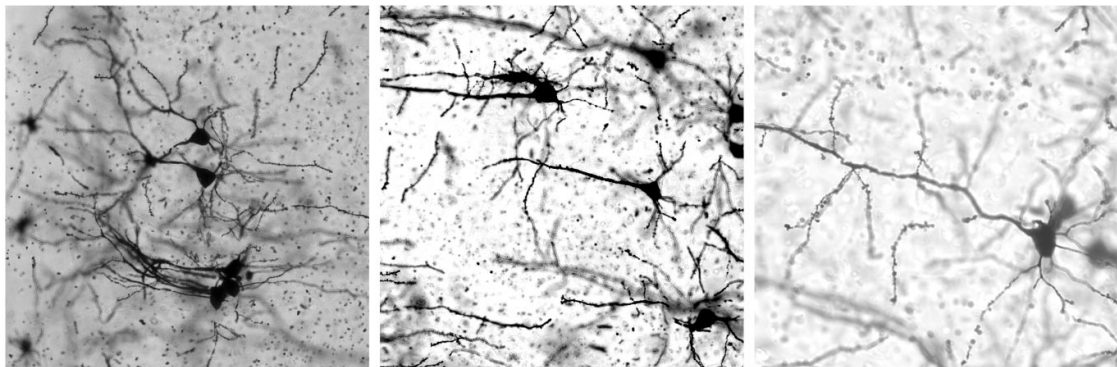
The most plausible explanation is random nucleation: a threshold concentration of the mercury-chromate complex must accumulate (nucleate) inside a cell before a stable precipitate seed forms from where it spreads across the entire cell. Which neurons cross that threshold appears close to random rather than being tied to a specific neurochemical identity, cell type, or activity state at the time of harvest. Because nucleation is a chance event with a low per-cell probability. The practical consequence is suprisingly fortunate rather than unfortunate; because if every neuron filled, the tissue would be an unresolvable black mass and individual arbors could never be



Beautiful Dendritic tree arborisations of a cortical pyramidal neuron



A stellate neuron beside a another neuron that has a long projection with spines all over its length visualized in a image stack



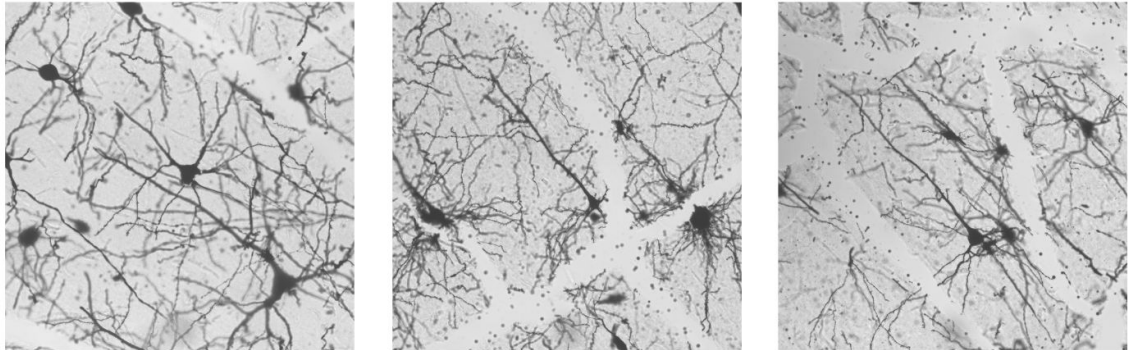
Brightfield micrographs of Golgi-Cox stained Cortical neurons showing progressively higher-magnification views of dendritic arborization.

Low-magnification (10x) field showing a cluster of impregnated pyramidal neurons with apical and basal dendritic trees

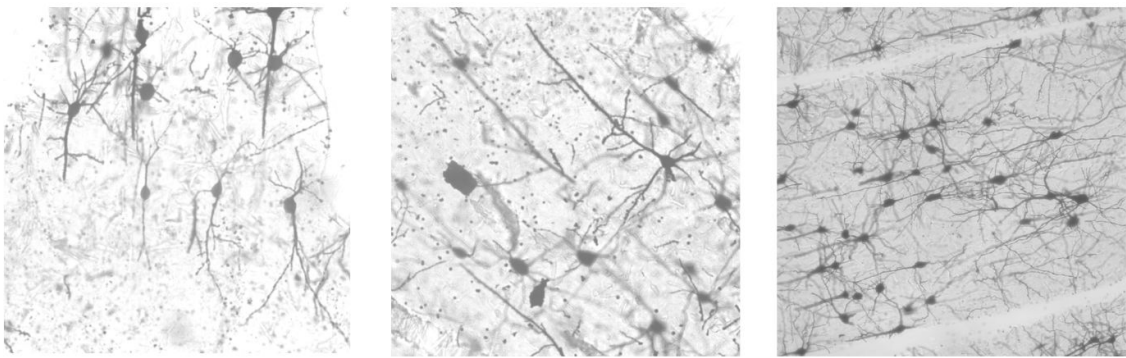
Intermediate-magnification (20x) field isolating two to three neurons, with the apical dendritic trunk and its proximal branch points clearly resolved.

High-magnification view of a single soma and its dendritic shaft, sufficient to resolve individual dendritic spines studding the branch

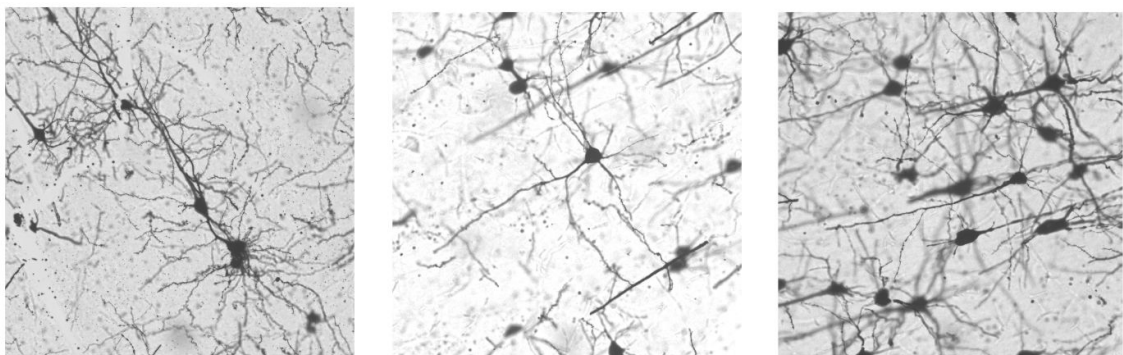
Figure 2. Visualizing Neuronal Morphology in the cortex



Cortical Neurons with long extended branching structures; also notice all the sections are tattered but still show the morphology of the neurons within preserved regions



Neurons from deeper in the cortex



Some more micrographs of neurons from hippocampus region; most parts were very densely packed these are some areas with lower density and isolated neurons

Figure 3. Visualization of Golgi-Cox stained neuronal morphology across brain regions.

traced. The 1–3% yield is what makes single-cell resolution possible.

A related open question was whether the physiological/electrical state of the neuron at the moment of harvest (resting potential, recent activity, membrane composition) determines which cells fill. We do not know if Golgi–Cox selectively labels neurons by activity state. We can however push selectivity of neuron vs. glia pharmacologically, but activity-dependent selectivity within the neuronal population is not established.

Light and metal instruments are avoided because both can catalyse premature reduction/oxidation of the chromate-mercury complex, causing diffuse background precipitation which we did observe in our sections as well to some extent.

5.2. Why do we take relatively thicker sections of around 150 μm ?

We deliberately take thicker sections than the normal (5–50 μm) as we want to preserve whole neurons. Thick slices (100–200 μm) keep complex 3D dendritic trees intact rather than slicing them into tiny fragments. The stain targets only 1–3% of cells, leaving the vast majority of the surrounding tissue completely transparent. This allows us to trace complete cell morphology and count dendritic spines across a single, continuous plane.

5.3. How can we preventing Tissue Tattering:

Hyper-osmotic shock during rapid 100% ethanol dehydration and aggressive lipid stripping by Xylene cause 150 μm sections to become brittle and fracture. To resolve this:

- Transition through a gradual ethanol gradient could be optimised.
- Xylene step should be done carefully or we could try to substitute Xylene with gentler clearing agents.
- We could increase slide coating to 5% gelatin with 0.5% chrom alum for better section adhesion.

5.4. Is there regional Variation in Impregnation?

We observed consistently darker, denser impregnation in hippocampal fields relative to cortex. Dense cytoarchitecture locally concentrates the mercury-chromate complex rather than allowing it to dissipate across a larger tissue volume, raising the effective reagent concentration available for intracellular nucleation at each cell. Cortical layers,

being more architecturally open, diffuse the complex more freely, reducing local concentration and producing lighter, more sparse filling. This suggests that a single fixed impregnation time in the protocol may not be universally optimal across brain regions within the same experiment.

How could this answer questions about Autism in the model?

Golgi-Cox staining enables precise quantification of altered dendritic spine density, regional impregnation pattern and arbor branching complexity. Comparing these against a control rat could hint at disease pathophysiology. Whether the changes are cell-type specific; pyramidal neurons versus interneurons, for instance, could imply very different circuit-level consequences.

Resolution Limits: Brightfield Z-stacking allows continuous 3D dendritic tracing, but resolving individual spine head morphology in 150 μm sections requires confocal or two-photon microscopy to eliminate the out-of-focus light that brightfield cannot reject at that depth.

6. CONCLUSION

The Golgi–Cox method, despite being around 150 years old and refined through multiple studies, remains the most reproducible standard for visualising single-neurone morphology and quantifying dendritic parameters at the individual cell level. Through this protocol we successfully isolated and identified pyramidal, stellate, and interneuron subtypes across the cortex and hippocampus to appreciate the beauty in the complexity of the brain.

Acknowledgement: I am deeply grateful to Dr. Krithika for her flexibility and support in allowing me to learn this protocol alongside my other projects. Sincere thanks to Project Associates Adrish, for his guidance on the tissue processing and with the epifluorescence microscope; and Abanti, for teaching me animal handling, dissection, and vibrotome sectioning. Finally, thank you to my fellow interns: Aashlesha, Sanaha, Aniket, and Shailesh—for their constant support and companionship, which made the learning process enjoyable.

Ethical Statement: This study follows ethical and academic integrity guidelines. It was conducted under the supervision of Project Associates at CHINTA.

7. REFERENCES

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